

CDN Cannon: Exploiting CDN Back-to-Origin Strategies for Amplification Attacks

CDN Cannon: Exploiting CDN Back-to-Origin Strategies for Amplification Attacks - Author not stated, [usenix.org](https://www.usenix.org).

- Published: date not stated
- Original: <<https://www.usenix.org/conference/usenixsecurity24/presentation/lin-ziyu>>
- Preserved from: <https://www.usenix.org/conference/usenixsecurity24/presentation/lin-ziyu> (live) on 2026-08-08
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CDN Cannon: Exploiting CDN Back-to-Origin Strategies for Amplification Attacks

Ziyu Lin, *Fuzhou University and Tsinghua University*; Zhiwei Lin, *Sichuan University and Tsinghua University*; Ximeng Liu, *Fuzhou University*; Jianjun Chen and Run Guo, *Tsinghua University*; Cheng Chen and Shaodong Xiao, *Fuzhou University*

Content Delivery Networks (CDNs) provide high availability, speed up content delivery, and safeguard against DDoS attacks for their hosting websites. To achieve the aforementioned objectives, CDN designs several 'back-to-origin' strategies that proactively pre-pull resources and modify HTTP requests and responses. However, our research reveals that these 'back-to-origin' strategies prioritize performance over security, which can lead to excessive consumption of the website's bandwidth.

We have proposed a new class of amplification attacks called **Back-to-Origin Amplification (BtOAmP)** Attacks. These attacks allow malicious attackers to exploit the 'back-to-origin' strategies, triggering the CDN to greedily demand more-than-necessary resources from websites, which finally blows the websites. We evaluated the feasibility and real-world impacts of 'BtOAmP' attacks on fourteen popular CDNs. With real-world threat evaluation, our attack threatens all mainstream websites hosted on CDNs. We responsibly disclosed the details of our attack to the affected CDN vendors and proposed possible mitigation solutions.

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BibTeX

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@inproceedings {298116, author = {Ziyu Lin and Zhiwei Lin and Ximeng Liu and Jianjun Chen and Run Guo and Cheng Chen and Shaodong Xiao}, title = {{CDN} Cannon: Exploiting {CDN} {Back-to-Origin} Strategies for Amplification Attacks}, booktitle = {33rd USENIX Security Symposium (USENIX Security 24)}, year = {2024}, isbn = {978-1-939133-44-1}, address = {Philadelphia, PA}, pages = {5717--5734}, url = {https://www.usenix.org/conference/usenixsecurity24/presentation/lin-ziyu}, publisher = {USENIX Association}, month = aug }
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